FPT UNIVERSITY
DATA SCIENCE METHODOLOGY

10-Step SQL & Python Analytics Pipeline

- **6,607 Student Records**
- **16 Attributes**
- **14 SQL Queries**
- **Statistical & ML Models**

Final Project Defense
Student Performance Factors

Subject: Data Science with Python & SQL (ADY201m)
Dataset: StudentPerformanceFactors.csv (6,607 rows × 16 columns)

Project Deliverables & Core Highlights

- 10-Step Methodology: Business Understanding -> SQL/Python Cleaning -> Modeling -> Deployment.
- Non-Destructive SQL Cleaning: CTE + ROW_NUMBER(), flagging Exam_Score = 101 as NULL.
- Comprehensive EDA: 7 high-resolution visualizations, IQR outlier detection, Pearson correlation.
- Statistical & Predictive Modeling: One-way ANOVA, T-tests, SQL Window Functions & Linear Regression.

01

AGENDA

Presentation Outline

01. Business Understanding & Approach

• Problem definition, 3 core questions, SQL + Python methodology.

02. Data Requirements & SQL Collection

• 6,607 records, 16 variables, rowid key generation, initial SQL queries.

03. Data Understanding & Preparation

• Handling missing values ('Unknown') and invalid score (101 -> NULL).

04. Exploratory Data Analysis (EDA)

• Target score distribution, IQR outliers, and correlation matrix.

05. Categorical & Demographic Analysis

• Violin/Box plots across family, school, peer, gender, and school type.

06. SQL Rule-Based Segmentation

• CASE WHEN study tiers, RANK() OVER window function, attendance groups.

07. ANOVA Testing & Linear Regression

• 8/10 significant categorical factors ($p < 0.001$) & regression R^2 = 60.8%–68.4%.

08. Evaluation, Deployment & Feedback

• SQL CTE + JOIN comparisons, CSV/DB deliverables, and recommendations.

Structured around the 10-Step Data Science Methodology with emphasis on Steps 4–8.

02

STEPS 1 TO 3 — METHODOLOGY

Business Understanding, Approach & Data Requirements

Step 1: Business Understanding

- Problem: Identify study habits and contextual factors linked to high/low Exam_Score.
- Q1: What is the average exam score (67.23) and data quality status?
- Q2: How do weekly study hours and attendance rates differentiate student scores?
- Q3: How do motivation, parental involvement/education, and extracurriculars interact?

Step 2: Analytic Approach (SQL + Python)

- SQL (SQLite): Quality checks, non-destructive cleaning (CTE + ROW_NUMBER), CASE WHEN segmentation, Window Functions (RANK, SUM OVER), and CTE + JOIN benchmarking.
- Python (Pandas, SciPy, Scikit-learn): Cross-verification of SQL results, IQR outlier analysis, One-way ANOVA / T-tests, and Multivariate Linear Regression.

Step 3: Data Requirements (StudentPerformanceFactors.csv — 6,607 Rows × 16 Columns)

- Primary Key: Raw CSV lacks a key column -> generated sequential student_id = rowid (1 to 6,607).
- 6 Numerical Variables: Exam_Score (target, 0–100), Hours_Studied (1–44h/wk), Attendance (60–100%), Previous_Scores (50–100), Sleep_Hours (4–10h), Physical_Activity (0–6h).
- 10 Categorical Variables: Parental_Involvement, Access_to_Resources, Motivation_Level, Family_Income, Teacher_Quality (78 missing), Peer_Influence, Parental_Education_Level (90 missing), Extracurricular_Activities, School_Type, Gender.

Dual SQL and Python implementation guarantees accuracy, interpretability, and reproducibility.

03

STEPS 4 TO 6 — DATA PREPARATION

SQL Data Collection, Quality Audit & Cleaning Pipeline

Steps 4 & 5: SQL Collection & Quality Audit

- Loaded 6,607 rows into SQLite table raw_students.
- Query 4.1 (Top 10 Attendance >= 90%): Detected student_id = 1526 with Exam_Score = 101.
- Query 4.2 (School_Type × Gender): Public (4,598 students, avg ~67.21) vs. Private (2,009 students, avg ~67.29) — 100% matched with Pandas.
- Audit: 0 duplicates | 78 NULL in Teacher_Quality | 90 NULL in Parental_Education | 1 score > 100.

Step 6: Non-Destructive Cleaning (SQL & Python)

- Created clean table students_clean_sql & DataFrame clean_py (raw data untouched).
- Rule 1 & 2: student_id = rowid (1..6607); lowercase snake_case columns & TRIM() text.
- Rule 3: Map missing categorical values (78 & 90) to 'Unknown' via COALESCE/NULLIF.
- Rule 4: Flag out-of-range Exam_Score = 101 as NULL (never fabricate scores).
- Rule 5: Deduplicate via ROW_NUMBER() OVER (PARTITION BY ...) = 1.

Before vs. After Cleaning Reconciliation Summary

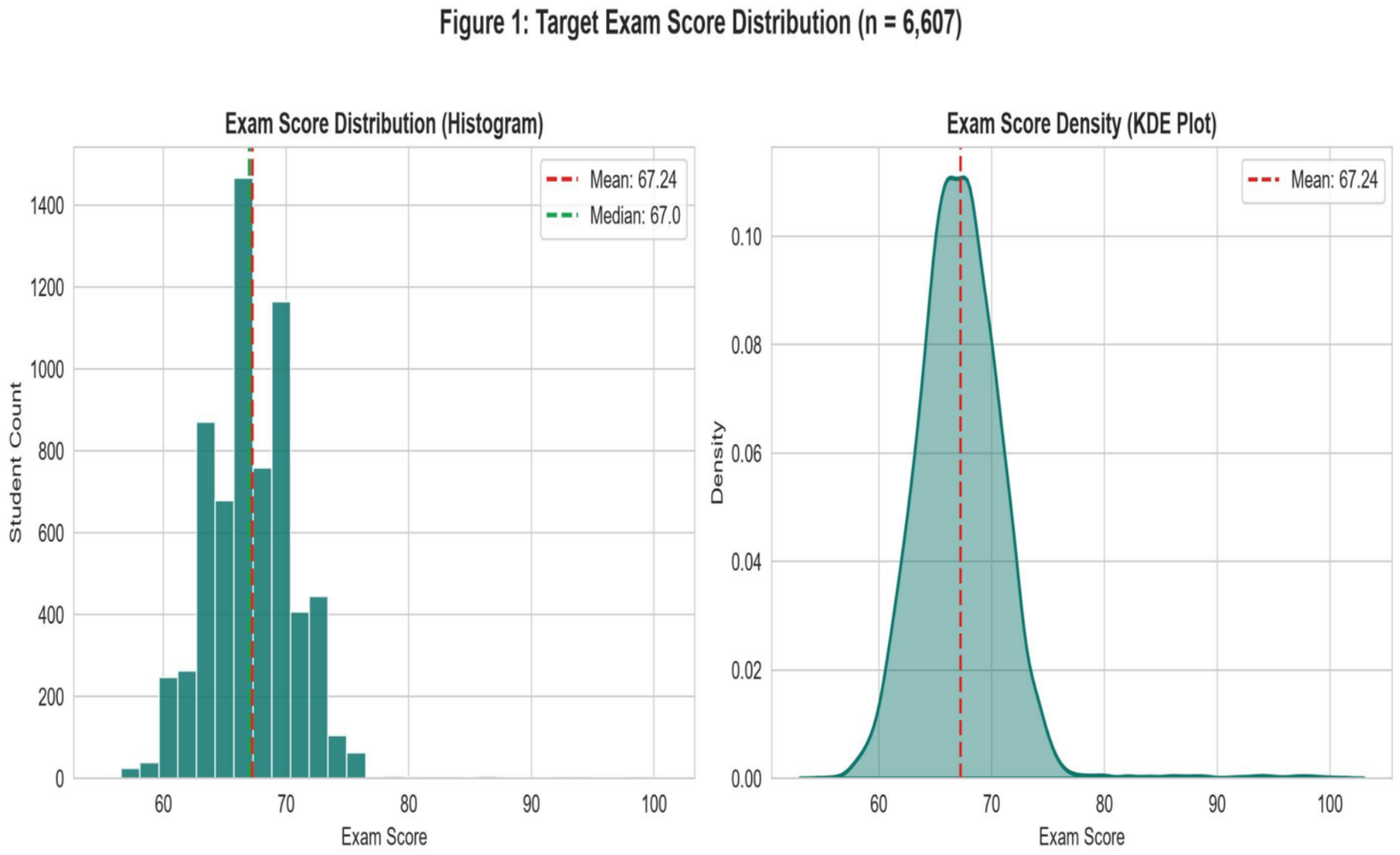
- Records: 6,607 Before -> 6,607 After (with 6,606 valid Exam_Score records and 1 NULL anomaly at student_id = 1526).
- Missing Categorical Cells: 168 total missing cells (78 Teacher_Quality + 90 Parental_Education_Level) -> 0 missing cells (labeled 'Unknown').
- Valid Mean Exam_Score: 67.236 (Raw with score 101) -> 67.231 (Cleaned SQL & Python, exact match to 8 decimal places).

Step 6 achieves 100% reconciliation between students_clean_sql and students_clean_python.

04

EXPLORATORY DATA ANALYSIS

Target Exam Score Distribution & Outliers



Distribution Summary

- Mean: 67.24 | Median: 67.0.
- Std Dev: 3.89 | IQR: 4.0 (Q1=65, Q3=69).
- Range: 55 to 100 (valid), 101 (flagged NULL).
- Skewness = 1.645 (right-skewed tail of top performers scoring 76–100).
- IQR Outlier Check:
  - Hours_Studied: 43 (0.65%)
  - Exam_Score: 104 (1.57%)
  - Other 4 numeric vars: 0%

Most students score between 65 and 69, with a positive tail of high-performing students.

05

EXPLORATORY DATA ANALYSIS

Key Numerical Drivers & Correlation Matrix

Figure 2: Relationship Between Study Hours, Attendance, and Exam Score

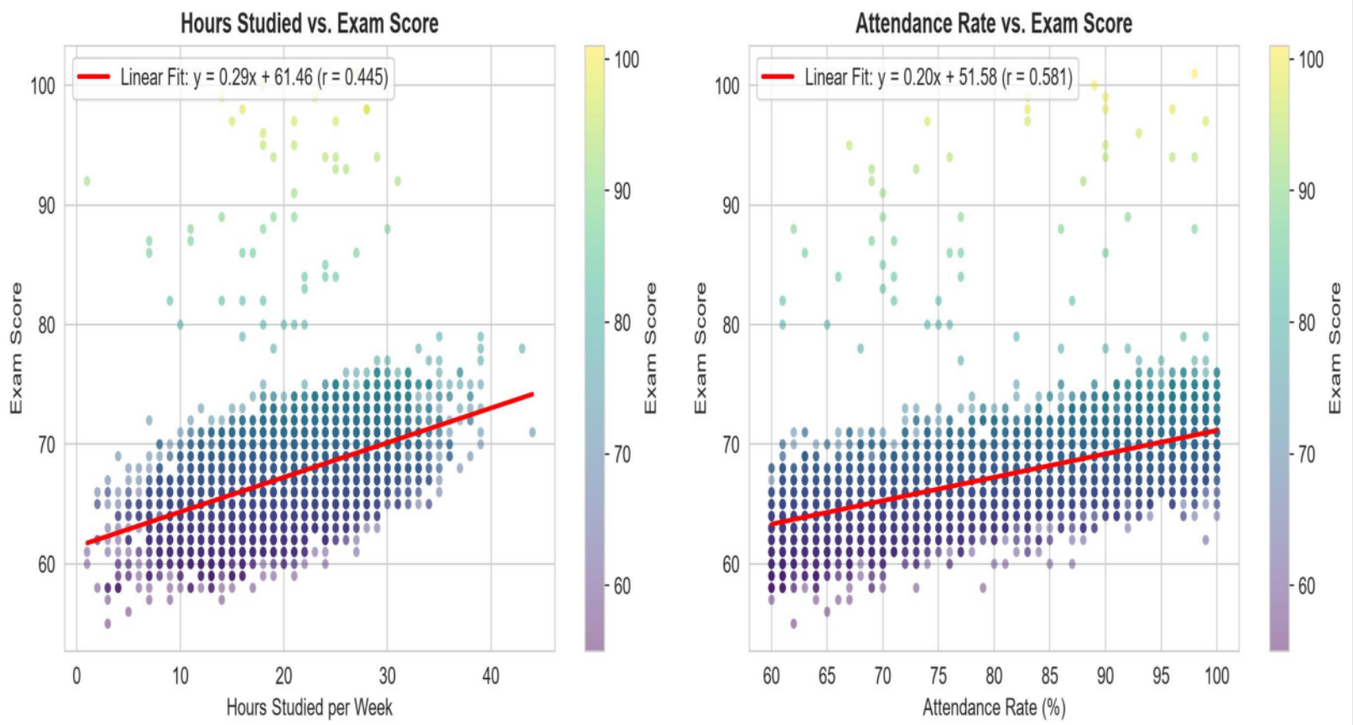
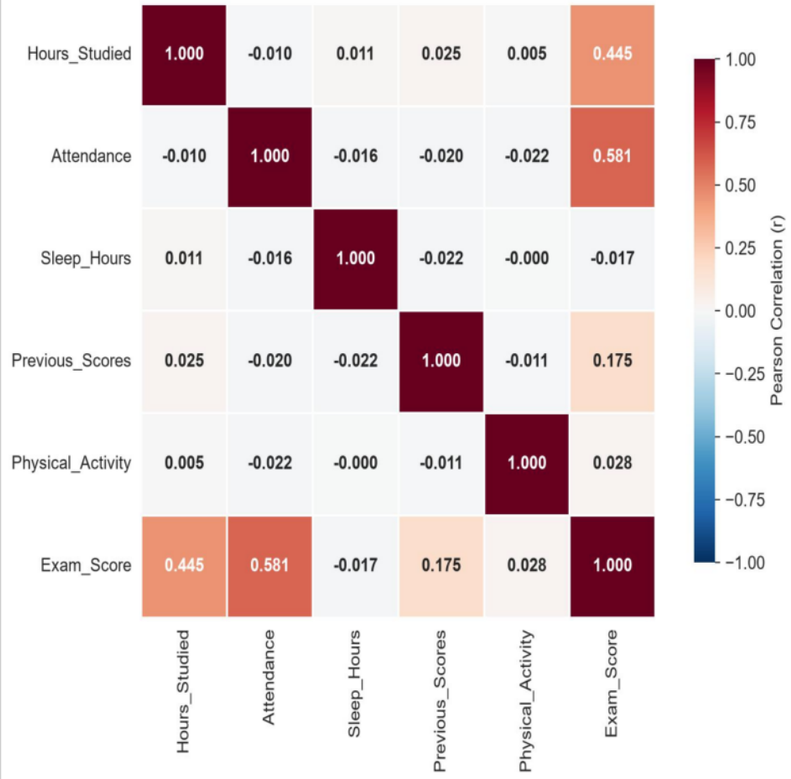


Figure 3: Pearson Correlation Matrix of Numerical Variables



Key Findings: Attendance (r = 0.581) and Hours_Studied (r = 0.445) are the primary numerical drivers; Previous_Scores (r = 0.175) is secondary.

Sleep_Hours (r = -0.017) and Physical_Activity (r = 0.028) show almost zero linear correlation with Exam_Score.

06

EXPLORATORY DATA ANALYSIS

Impact of Categorical & Demographic Factors

Figure 4: Impact of Key Categorical Factors on Exam Score (ANOVA p < 0.001)

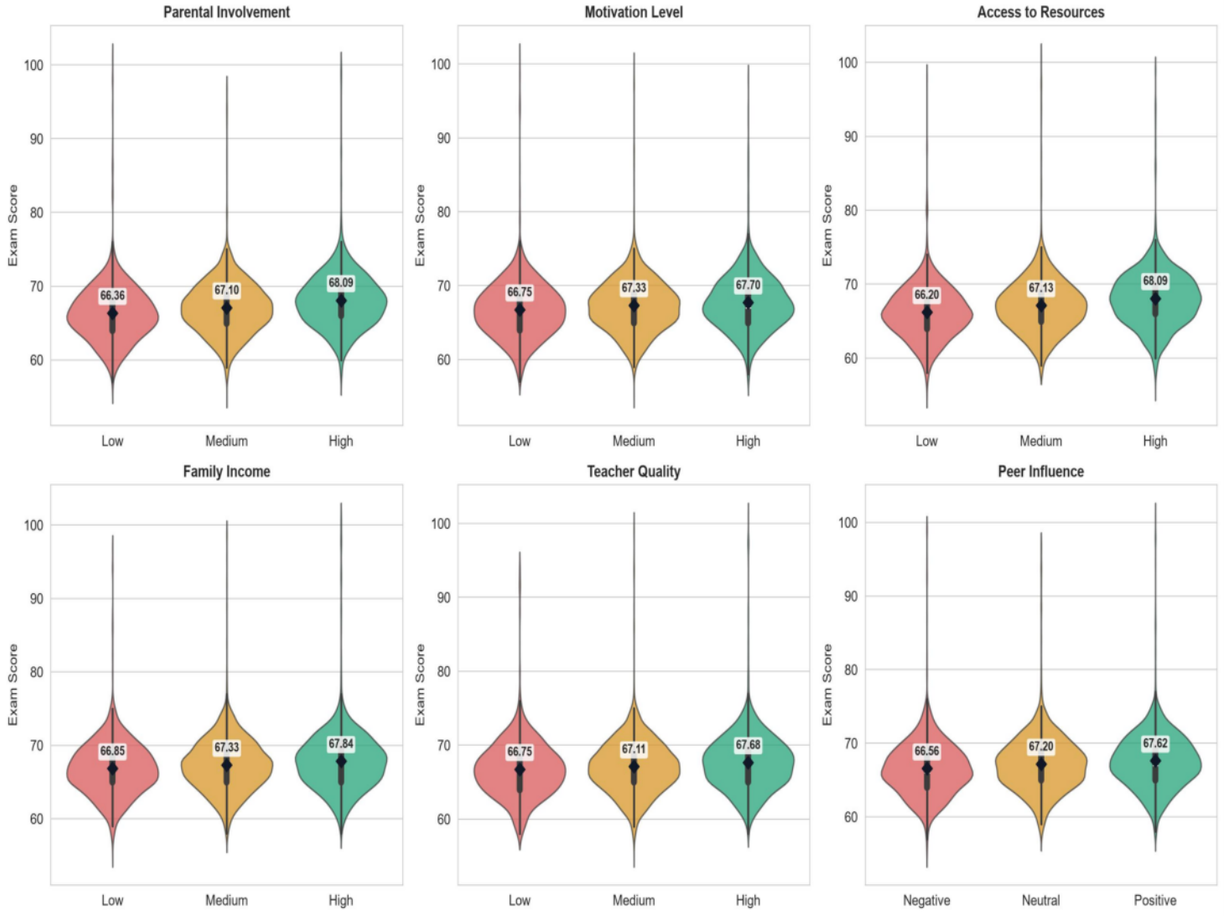
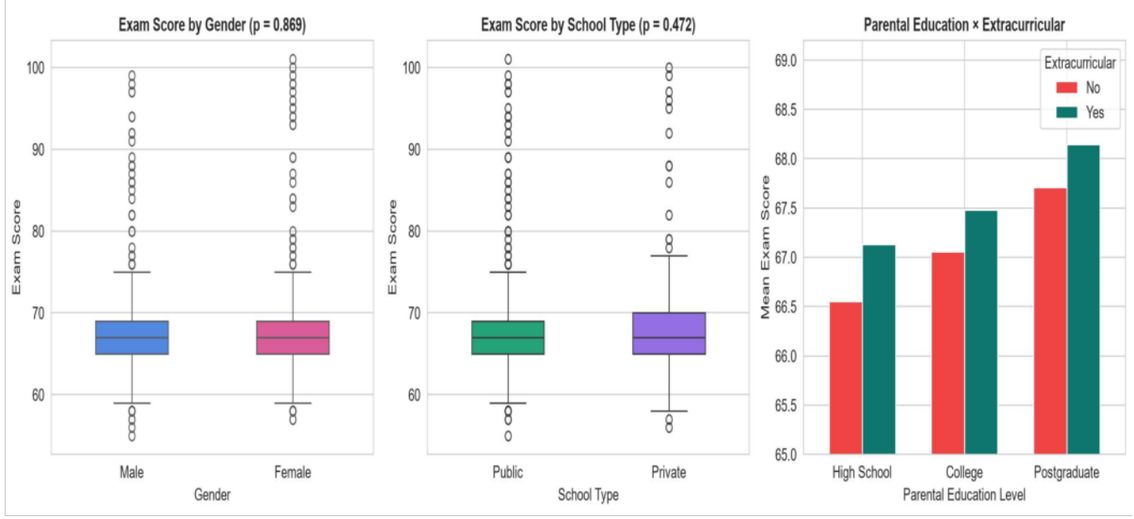


Figure 5: Score Comparison Across Demographics and Extracurricular Participation



Significant vs. Non-Significant Categorical Factors

- Strong Positive Impact (p < 0.001): Access_to_Resources (+1.89 pts), Parental_Involvement (+1.73 pts), Parental_Education (+1.08 pts), Peer_Influence (+1.06 pts), Family_Income (+0.99 pts), Motivation_Level (+0.95 pts), Teacher_Quality (+0.93 pts), Extracurricular (+0.51 pts).

- No Significant Impact: Gender (p = 0.869) & School_Type (p = 0.472).

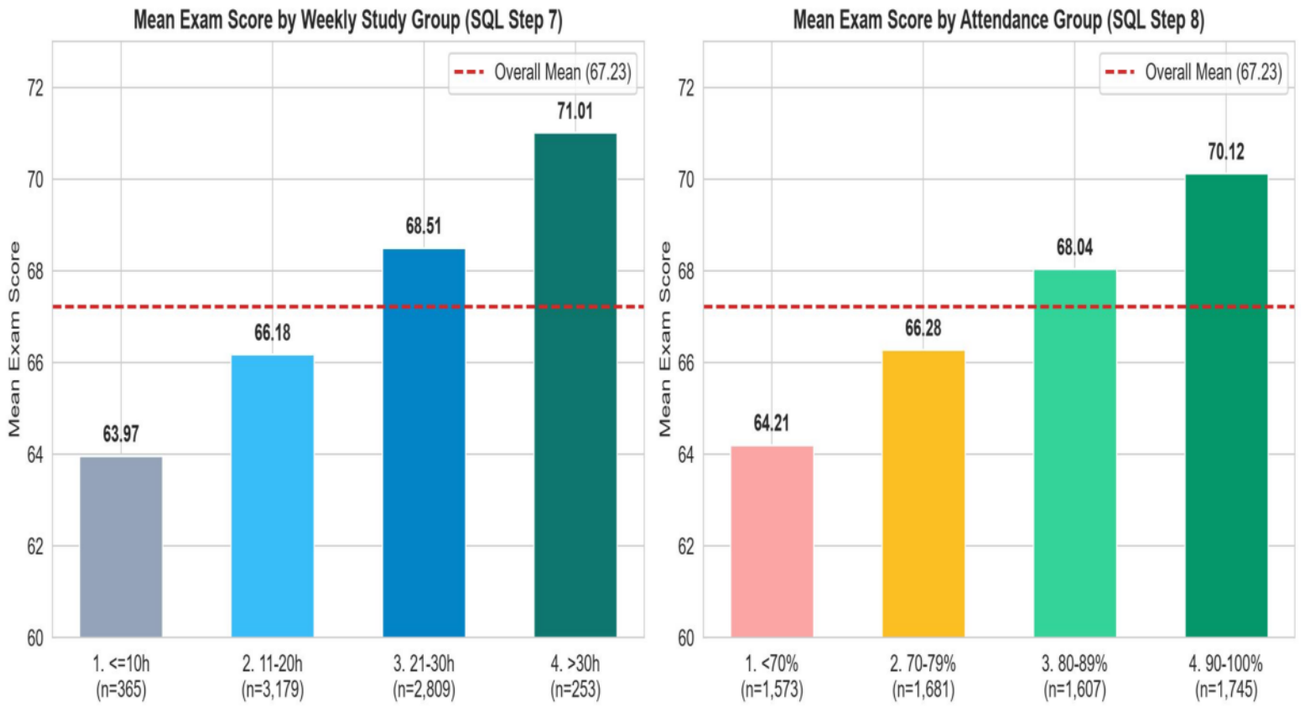
Family involvement, learning resources, and positive peers consistently boost academic performance.

07

STEPS 7 & 8 — SQL MODELING

SQL Rule-Based Segmentation: Study Hours & Attendance

Figure 6: SQL Rule-Based Segmentation by Study Hours and Attendance Rate



SQL Window Function Results

- Study Groups (Step 7):
  - <=10h (n=365): 63.97 (-3.26, Rank 4)
  - 11-20h (n=3,179): 66.18 (-1.05, Rank 3)
  - 21-30h (n=2,809): 68.51 (+1.28, Rank 2)
  - >30h (n=253): 71.01 (+3.78, Rank 1)

- Attendance Groups (Step 8):
  - <70% (n=1,573): 64.21
  - 70-79% (n=1,681): 66.28
  - 80-89% (n=1,607): 68.04
  - 90-100% (n=1,746): 70.12

SQL Technique: Used WITH grouped AS (CASE WHEN ...), RANK() OVER (ORDER BY avg_exam_score DESC), and weighted diff vs. overall average.

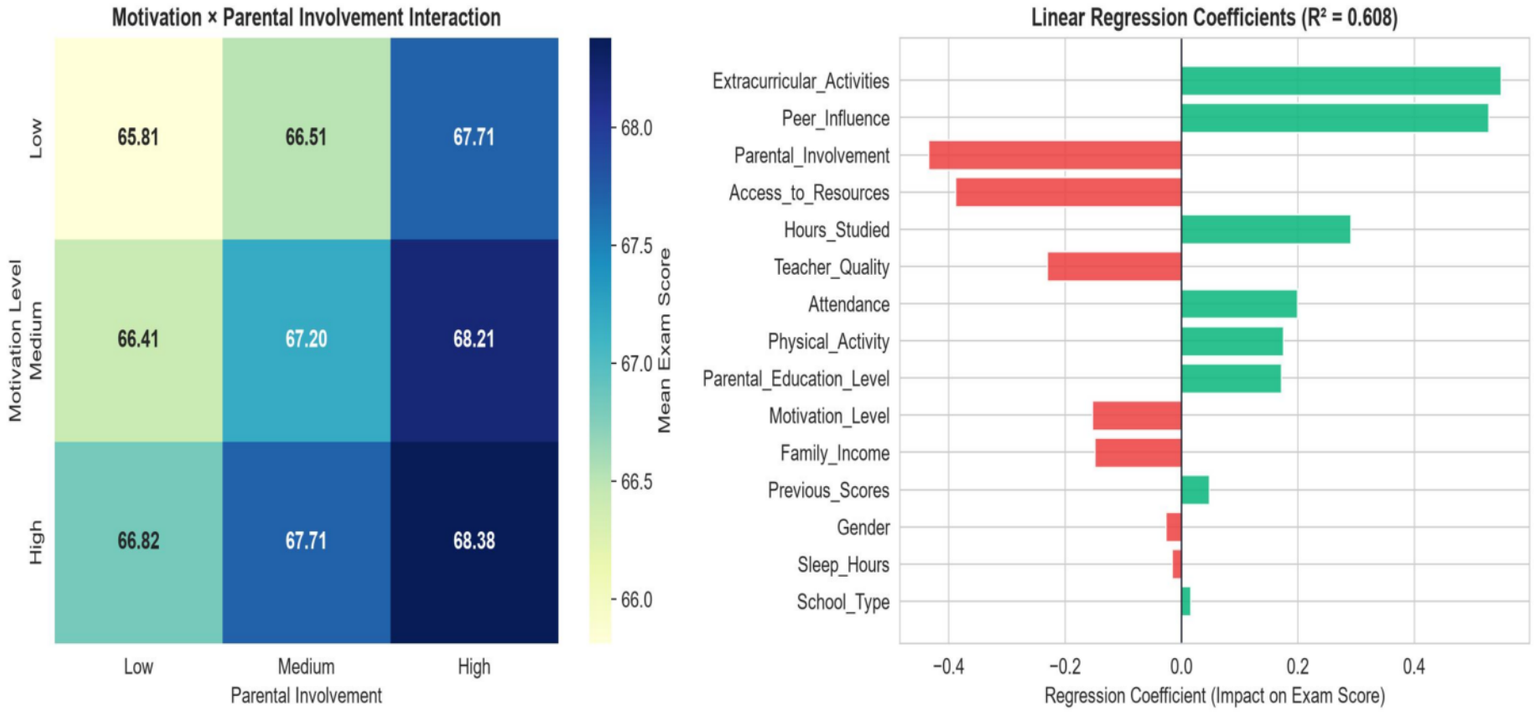
Moving from <=10h to >30h of weekly study raises mean exam scores by +7.04 points.

08

STATISTICAL & ML MODELING

Interaction Matrix & Linear Regression Model

Figure 7: Multi-Factor Interaction Matrix and Regression Feature Importance



Model Findings

- Motivation × Parental Involvement:
  - Low × Low: 65.81
  - Medium × Medium: 67.20
  - High × High: 68.38 (+2.57 pts)

- Linear Regression:
  - Label-Encoded R^2 = 0.608 (60.8% variance)
  - One-Hot R^2 = 0.684 (68.4% variance)
  - Hold-out Test MAE = 1.31 pts, RMSE = 2.62 pts.

Synergistic Effect: Behavioral effort (study hours, attendance) interacts positively with family involvement and motivation.

Multivariate Linear Regression confirms that 13 statistically significant factors jointly explain up to 68.4% of score variance.

09

STEPS 8 TO 10 — CONCLUSION

Evaluation (JOINs), Deployment & Recommendations

Step 8: Advanced SQL CTE + JOIN Evaluation

- CROSS JOIN (Parental Education × Extracurricular vs. Overall Mean 67.23):
  - Postgraduate + Yes (n=774): 68.15 (+0.92)
  - College + Yes (n=1,208): 67.48 (+0.25)
  - High School + No (n=1,317): 66.55 (-0.68)

- INNER JOIN (Top Students Exceeding Motivation Group Mean):
  - student_id = 95 (Medium, Score = 100, +32.67 vs group)
  - student_id = 2426 (Low, Score = 99, +32.27 vs group)

Step 9: Deployment Deliverables

- SQLite Database: student_analysis.db containing raw_students, students_clean_sql, and students_clean_python.

- Standalone SQL Script: TruyVan_LamSach_PhanTich.sql (14 reproducible SQLite queries).

- Exported Clean CSVs (UTF-8-SIG with lock-safe fallback):
  - student_performance_cleaned_sql.csv
  - student_performance_cleaned_python.csv
  - analysis_study_groups.csv

Step 10: Feedback & Strategic Recommendations for Educational Institutions

- 1. Data Governance: Verify anomalous record student_id = 1526 (Exam_Score = 101) with the exam board; enforce mandatory fields for Teacher_Quality & Parental_Education.

- 2. Early Academic Warning: Automatically alert advisors when student Attendance drops below 75% or weekly self-study falls below 10 hours.

- 3. Holistic Student Support: Expand learning resources for low-income students, strengthen parent-school communication, and promote extracurricular clubs.

Thank You for Your Attention! — End-to-End 10-Step Data Science Pipeline with SQL & Python